

Acoustic Communication – Diver Alert System

Abstract

This project demonstrates a simple underwater wireless communication system using ultrasonic waves. The system is designed to provide diver alerts and underwater signaling, ensuring safety and effective communication in aquatic environments.

Core Technology

- **Modulation Used:** On–Off Keying (OOK) modulation, where the presence or absence of ultrasonic pulses represents binary data.
- **Hardware:** Microcontrollers (Tx/Rx), ultrasonic transducers (40 kHz), push button, buzzer, LCD.
- **Software:** Embedded C/Arduino IDE for microcontroller programming, signal detection and thresholding routines.

Novelty

- Low-cost prototype for underwater signaling.
- Combines audio (buzzer) and visual (LCD) alerts for redundancy.
- Demonstrates practical diver communication without complex RF equipment.

Limitations

- Short communication range due to ultrasonic attenuation.
- Limited to binary signaling (no complex data transfer).
- Sensitive to environmental noise and multipath effects.

Feasibility

- Highly feasible for diver alert and aquatic safety systems.
- Extendable with advanced modulation (FSK/PSK) and error correction.
- Scalable to aquaculture monitoring and shallow marine applications.

Conclusion

- The prototype represents a basic yet effective underwater communication system. By using ultrasonic waves and OOK modulation, it enables reliable signaling between divers and monitoring stations, contributing to enhanced safety in aquatic environments.

